



Wojood^{Relations}: Arabic Relation Extraction Corpus and Modeling



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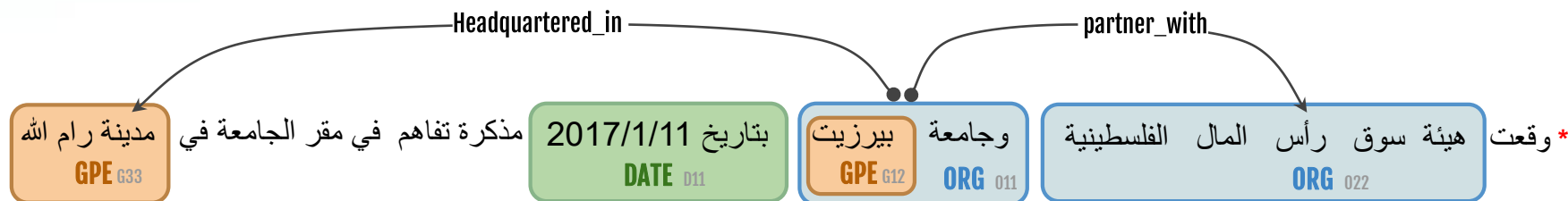
المساعدات الآلية

Motivation

- ❖ Relation extraction (RE) is a **core task in natural language processing**, crucial for semantic understanding, knowledge graph construction, and enhancing downstream applications.
- ❖ A notable **gap in the availability of annotated corpora** for this purpose, especially for under-resourced languages like Arabic.

Relation Extraction

- ❖ **Input** : a sentence with predicted named entities.
- ❖ **Goal** : Extract relations between pairs of entities.



Subject	Object	Relation	Relation Aware Explanation
O22	O11	Partner_with	لأن الجملة تصف توقيع مذكرة تفاهم، وهي صيغة تدل عادةً على ...
O11	G33	Headquartered_in	كلمة "مقر الكلية" تعني المكان الأساسي الذي تعمل منه الكلية،...

*<O2> The Palestinian Capital Market Authority </O22> and <O11> <G12> Birzeit </G12> University </O11> signed a memorandum of understanding on <D11> January 11, 2017 </D11> at the university's headquarters in <G33> Ramallah </G33>.

Our contribution

- 
- ❖ **Wojood^{Relations}** corpus: large Arabic RE corpus with **550K tokens** and **40 annotated relation types**.
 - ❖ Baselines computed using **BERT-based**, **GNN-based**, and **LLM-based models** in both pipelined and joint extraction settings.
 - ❖ **Novel methodologies**, **NLI-RE**, a relation extraction framework, and **GPT-Joint**, a method for **few-shot joint entity and relation extraction**.
 - ❖ **A comparative analysis among different methods**, highlighting key challenges in Arabic relation extraction.
 - ❖ **End-to-End System** for **Arabic relation extraction**.

Wojood^{*Relations*} Corpus

Wojood^{Relations} Corpus

- ❖ Wojood is a rich Arabic nested named entity corpus with **550k tokens**, contains **75K annotated entities**. (Jarrar et al., 2022b)
- ❖ Extended the existing Wojood corpus to construct a relation extraction corpus.
- ❖ Each relation is defined formally, with specific **Domain** and **Range**.

Methodology	Dataset	Sentences	Rel. Types	Entity Types	Triplets	NER
Distant Supervision	SRED ^{FM}	46.6M [§]	393	13	3.3M	Flat
	SMiLAR	1.1M [§]	9	-	9K	Flat
Human Annotated	ACE05	30.9K [§]	6	5	4.7K	Flat
	Wojood ^{Hadath}	33K	3	21	2.8K	Nested
	RED ^{FM}	732	32	13	1.8K	Flat
	Our <i>Wojood^{Relations}</i>	<u>33K</u>	<u>40</u>	<u>21</u>	<u>14K</u>	Nested

Relationship Types

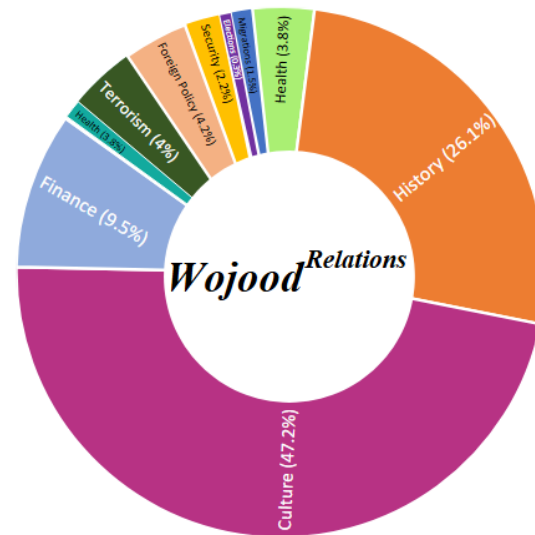
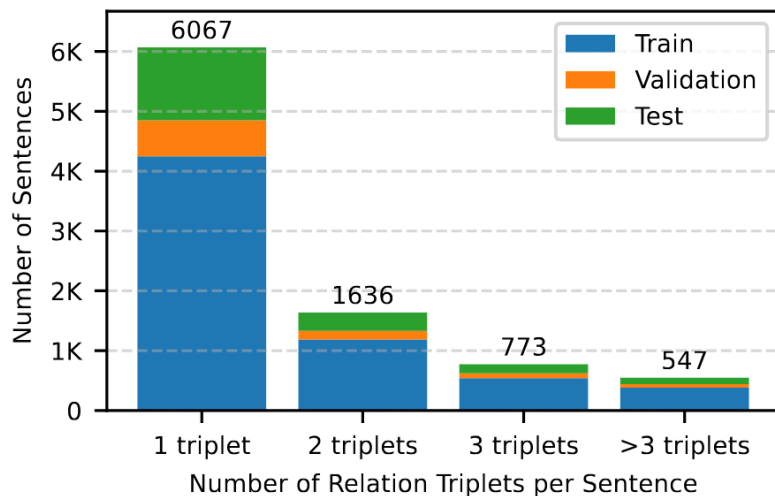
We annotated Woood with 40 relation types. The selection of these relations based on the following considerations:

- ❖ Frequent occurrence in Arabic texts.
- ❖ **Alignment with established ontologies** knowledge bases, which maintains semantic consistency and enables **interoperability**.

Relations		
birth_date	has_competitor	leader_of
birth_place	has_conflict_with	lives_in
branch_count	has_currency	located_in
builder_of	has_occupation	manager_of
capital_of	has_parent	manufacturer_of
death_date	has_population	member_of
employee_of	has_property	nearby
employs	has_relative	official_language
found_on	has_revenue	owner_of
founder_of	has_sibling	partner_with
geopolitical_division	has_spouse	president_of
has_alternate_name	headquartered_in	student_affiliation
has_area	inventor_of	subsidiary
has_border_with		

Corpus Statistics

❖ *Wojood^{Relations}* corpus includes **14, 689** relation triplets.



Annotation Guidelines

- ❖ Relations are recognized only **within the same sentence**.
- ❖ **Annotate** a relation only **if context explicitly indicates** one.
- ❖ **A nested structure does not imply a relation automatically.**
[“Beirut University” ORG: Beirut University, nested GPE: Beirut.]
- ❖ **Some relations are contextually inferred**, especially in governmental names.
[“Hebron Chamber of Commerce and Industry”]
- ❖ When an entity appears multiple times, **annotate only the first, full mention**.
[The Board of Directors of the Capital Market Authority... appointed Mr. Murad Jadba as the Director General of the Authority.]

Inter-annotator Agreement

- ❖ **Randomly selecting 10%** of annotations for evaluation:
 - Randomly sample **2% of relations from each domain** (total 10%).
 - For each relation, which may span multiple domain–range pairs, we randomly sampled **10% of candidate sentences per domain–range**.
- ❖ Inter-annotator Agreement **0.92 Cohen's kappa** and **89.5 F1 score**.

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Relation Extraction Modeling

Relation Extraction Baselines

- ❖ RE is commonly addressed through different methods:
 - Graph-based methods to capture structural dependencies.
 - Supervised learning with pretrained language models.
 - Generative modeling using LLMs.
- ❖ **Baselines:** existing graph-based methods and RE LLM-based methods.
- ❖ Proposed Methods: **NLI-RE** as supervised learning, and **GPT-Joint** as LLM-based joint extraction.

Graph-Based Relation Extraction

- ❖ RE can be formulated as graph reasoning using heterogeneous graph neural networks [52].
- ❖ Each sentence represents tokens, entities, and relation types as nodes in a graph and performs message passing to update node embedding. Final node representations are used to predict relations.
- ❖ **Dataset:** Rel^{Graph} dataset from *WojoodRelations* splits. The final dataset includes 13,211 instances: 8,819 train, 1,500 validation, and 2,892 test.

LLM-Based Relation Extraction

- ❖ Reformulate RE as a sequence generation task using in-context learning.
- ❖ GPT-RE [53] prompts GPT with few-shot examples retrieved via entity-aware semantic embeddings, and augments them with reasoning based on gold labels.
- ❖ **Dataset:** Rel^{GPT} instruction dataset from *WojoodRelations* splits. The final dataset includes 13,211 instances: 8,819 train, 1,500 validation, and 2,892 test.

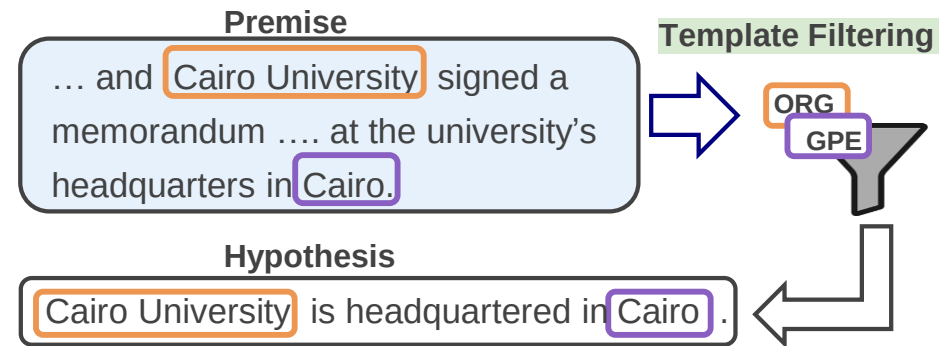
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NLI-Based Relation Extraction

❖ **NLI-RE** extend [8], by framing relation extraction as a **binary Natural Language Inference (NLI)** problem with **relation-aware templates**.

- 1) Relation Templates: Define a relation-aware template for each relation type, constrained by domain and range types.
- 2) Premise–Hypothesis Generation: Treat the sentence as the premise and generate hypotheses for each entity pair using matching templates.
- 3) Labeling: Assign True if the relation exists in `WojoodRelations`, otherwise False.
- 4) Classification: Encode premise–hypothesis pairs with BERT and classify relations.



NLI-Based Relation Extraction

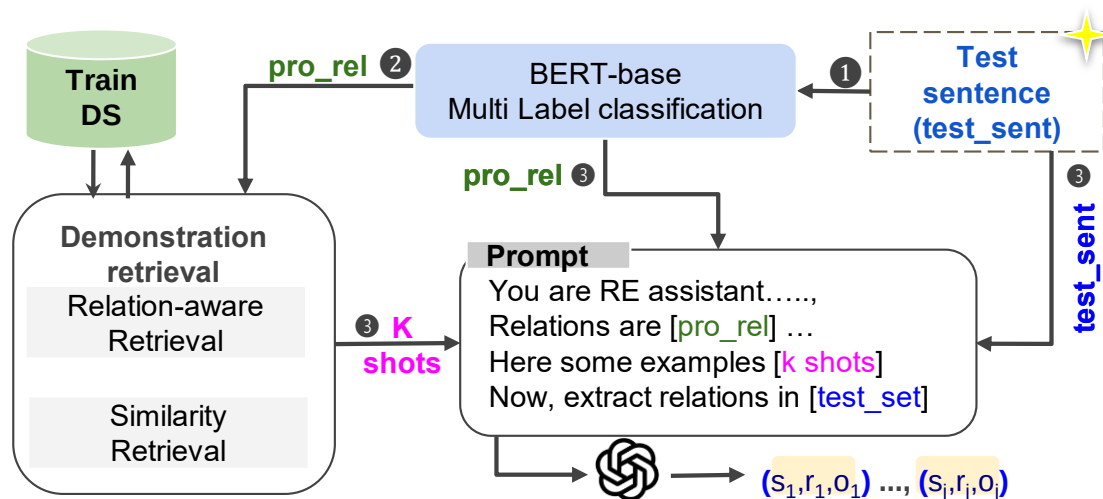
- ❖ **Dataset:** Construct Rel^{NLI} by converting each sentence–entity pair in *WojoodRelations* into corresponding premise–hypothesis pairs using relation-aware templates. The dataset statistics are

Split	Number of Sentences	Positive	Negative
Train	20,300	10,319	9,981
Validation	3,106	1,474	1,632
Test	5,971	3,892	2,079
Total	29,377	15,685	13,692

LLM-Based Joint Entity and Relation Extraction

❖ Three-step process:

- 1) Predicting candidate relation types.
- 2) Retrieving relationally and semantically similar examples.
- 3) Generating relation triplets via LLM prompting.



Results

❖ **Relation extraction** results on *Wojood*^{Relations} test set.

Model	Micro P	Micro R	Micro F1	Macro P	Macro R	Macro F1
NLI-RE (supervised)	88.91	88.65	88.61	89.00	88.50	<u>88.58</u>
RIFRE (supervised)	93.29	93.28	<u>92.89</u>	54.36	54.34	52.05
GPT-RE (LLM)	89.25	83.66	85.78	69.20	68.75	63.55

Results

❖ Joint entity and relation extraction results on *Wojood^{Relations}* test set.

Model	Zero-Shot			Few-Shot		
	P	R	F1	P	R	F1
Llama3-70b	35.00	28.43	31.37	59.06	67.58	63.03
deepseek-distill-llama	27.24	18.50	22.03	74.45	61.16	67.15
gpt-4o (LLM relation proposal)	43.30	29.83	<u>35.32</u>	80.39	60.20	68.85
gpt-4o (BERT relation proposal)	-	-	-	65.92	62.90	64.38
deepseek-reasoner	35.35	29.02	31.87	76.20	69.56	<u>72.73</u>

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A comparative Analysis Among Methods

Discussion

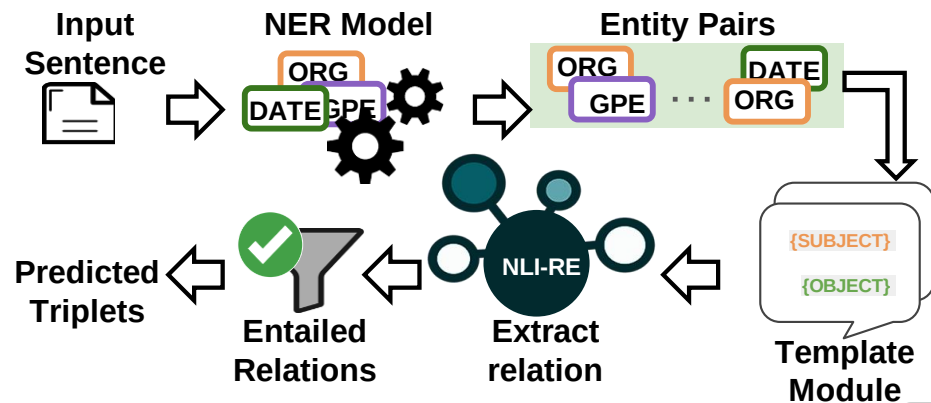
- ❖ **Overlapping and implicit relations are challenging**; most models predict only one relation per sentence.
- ❖ **NLI-RE**: Strong at inferring both explicit and implicit relations using contextual cues.
- ❖ GPT-RE outperforms RIFRE, but its dependence on surface-level patterns from retrieved examples limits its ability to generalize to less obvious relations.

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End to End System

- ❖ Develop an end-to-end relation extraction system as part of the SinaTools.
- ❖ Adopt NLI-RE because of its strong performance and can be implemented with lighter and faster models such as BERT.
- ❖ The complete workflow include:
 - NER Module.
 - Template Module.
 - RE Module.



Conclusion

- ❖ Introduced *WojoodRelations*, the largest Arabic RE corpus with 40 relation types and high-quality annotations.
- ❖ **Benchmarking:** Supervised models outperform LLMs on this corpus.
- ❖ **Challenges:** Detecting implicit and rare relations remains difficult.
- ❖ **Future work:** Improve generalization to long-tail relations and better leverage contextual information for Arabic RE.

Thank You

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